

Submission Summary

Conference Name

IEEE Chandigarh Subsection International Conference

Track Name

Artificial Intelligence and Data Science

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1478

Paper Title

From Chatbots to Agentic AI: A Comparative Analysis of Large Action Models in ITSM

Abstract

Traditional enterprise IT support is burdened by high-volume, low-complexity requests that exceed human capacity and overwhelm rigid, rule-based chatbots. To address the critical “automation gap” in modern enterprise architectures, this paper investigates the implementation of Salesforce Agentforce to transition toward an “Agentic Enterprise.” Utilizing a Design Science Research (DSR) methodology, an autonomous digital worker powered by the Atlas Reasoning Engine was deployed. The system utilizes a Reason-Act-Observe (ReAct) cycle to process complex, multi-stage objectives, with the Einstein Trust Layer ensuring data security and grounding. System performance was validated through controlled experiments using a categorized dataset of ITSM (Information Technology Service Management) tickets. To determine the performance of our proposed agentic architecture, we subjected it to a comprehensive evaluation, benchmarking it against two widely recognized industry systems: a conventional rule-based system and a standard Retrieval Augmented Generation (RAG) model. The results were definitive ($p < 0.001$), with the Agentforce model significantly outperforming these baselines. Specifically, the system achieved 91% accuracy in identifying user intent and successfully resolved 74% ($\pm 3.2\%$) of cases autonomously, marking a major leap forward from legacy technology. Despite the modest scale of the synthetic data, consistent empirical testing and detailed variance analysis offer compelling evidence for the efficacy of this architectural transition. This evidence is further supported by its alignment with current industry standards. Analysis of the agent’s internal logic confirmed its ability to handle erratic user behaviours, a factor that led directly to a significant reduction in Mean Time to Resolution (MTTR). Ultimately, these findings establish a validated framework for scalable enterprise automation, signalling a paradigm shift in how human and artificial intelligence collaborate within the IT sector.

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